INNOPOLIS UNIVERSITY

Bachelor of Science in Computer Science

ViKi: Video-to-Kinematics for Object-Centric Retargeting of Human Manipulation Demonstrations to Robot Manipulators

Bachelor's Graduation Thesis

Author: **Artyom [Surname]**

Supervisor: **[Full Name]**

Group: **[group]**

Innopolis, Russia

2026

Abstract

ViKi is a research pipeline for converting passively recorded, multi-view RGB-D demonstrations of tabletop manipulation into object-centric hand motion and candidate trajectories for a robot in simulation. Two Azure Kinect DK cameras capture a human moving a 40 mm blue cube from a red support cube to a green target cube. Hand and object perception feed an object-aware retargeter for an old-generation UR10 with a two-finger gripper. MuJoCo then executes and screens the candidate motion under contact dynamics. This separates a geometric trajectory proposal from physical success: an inverse-kinematics solution alone is not a successful grasp.

A local 78-episode cohort was audited for capture, hand extraction, object reconstruction and retargeting, and a bounded simulator study tested layout perturbations and a learned low-dimensional correction. These measurements came from one demonstrator and a largely fixed scene; the original local recordings and derived artifacts were later lost and must be regenerated before reproducible submission. In a separate historical HO-Cap pilot, two 120-frame sequences gave 10.50–11.67 mm hand-joint MPJPE against published semi-automatic MANO annotations for a selected camera pair and all eight views, with 2,520/2,520 matched joints per comparison. The pair was selected on the same sequences and the annotations do not measure the local Kinect rig. No physical-robot experiment, reinforcement-learning result, visual policy or Real-to-Sim-to-Real transfer is claimed.

Keywords: imitation learning; motion retargeting; inverse kinematics; RGB-D calibration; object-centric representation; robot learning from human video.

Аннотация

ViKi — исследовательский пайплайн преобразования пассивно записанных многокамерных RGB-D демонстраций настольной манипуляции в объектно-центричное движение кисти и кандидаты траекторий робота для симуляции. Две камеры Azure Kinect DK записывают перенос синего кубика с красного опорного кубика на зелёный. Оценка позы кисти и объекта передаётся объектно-ориентированному ретаргетингу на UR10 с двухпалым захватом; затем MuJoCo проверяет кандидатную траекторию с учётом контактов. Геометрически допустимое решение обратной кинематики само по себе не доказывает успешного захвата.

Локальный набор из 78 эпизодов был проанализирован по этапам записи, восстановления кисти и объекта, ретаргетинга и ограниченных экспериментов в симуляции. Данные получены одним демонстратором при почти фиксированной расстановке; исходные локальные записи и производные артефакты впоследствии были утрачены и требуют повторного сбора для воспроизводимой публикации. Отдельный исторический пилот на двух 120-кадровых последовательностях HO-Cap дал ошибку суставов кисти 10,50–11,67 мм относительно опубликованной полуавтоматической разметки MANO при покрытии 2520/2520 суставов для выбранной пары и всех восьми камер. Пара выбрана на тех же данных; результат не измеряет точность локального рига Kinect. Работа не заявляет испытаний на физическом роботе, обучения с подкреплением, визуальной политики или переноса Real-to-Sim-to-Real.

Ключевые слова: обучение с имитацией; ретаргетинг движения; обратная кинематика; калибровка RGB-D; объектно-центричное представление; обучение роботов по видео человека.

Acknowledgements

[Placeholder.] The author thanks **[supervisor]** for supervision and guidance throughout this work, and the Innopolis Robotics Society for maintaining a parallel development fork of the system, whose independent implementation attempts informed several of the architectural decisions reported here.

Contents

Abstract **i**

Аннотация **ii**

Acknowledgements **iii**

1 Introduction **1**

1.1 Background 1
1.2 Problem Statement 1
1.3 Research Gap 2
1.4 Research Questions and Hypotheses 2
1.5 Contributions 3
1.6 Thesis Structure 3

2 Literature Review **4**

2.1 Preamble 4
2.2 Representing the Human Hand 4
2.2.1 Full parametric reconstruction 4
2.2.2 Minimal representation: wrist pose and gripper state 5
2.2.3 Egocentric and monocular capture 5
2.3 Retargeting Formulations 6
2.3.1 The single cost functional 6

2.3.2 The sequential pattern and its critique . . . 6
2.3.3 Offline pipelines and their retargeting operators . . . 6
2.3.4 Solvers . . . 7
2.4 Task Representation . . . 7
2.5 Multi-Camera RGB-D Calibration, Fusion and Synchronisation . . . 8
2.5.1 Fiducial choice . . . 8
2.5.2 Reported accuracy of multi-Azure-Kinect rigs . . . 8
2.5.3 Depth fusion . . . 9
2.5.4 Cross-device synchronisation . . . 9
2.5.5 Hand–eye calibration . . . 9
2.6 Dataset Generation and Downstream Policies . . . 10
2.7 Positioning of the Proposed System . . . 10
2.8 Conclusion of the Review . . . 11

3 Design and Methodology **12**
3.1 System Overview . . . 12
3.2 Hardware and Workspace . . . 13
3.3 Calibration . . . 13
3.3.1 Intrinsics and extrinsics . . . 13
3.3.2 Robot registration . . . 14
3.4 Synchronisation . . . 14
3.5 Perception and Multi-View Reconstruction . . . 15
3.5.1 Landmark detection and lifting . . . 15
3.5.2 Multi-view triangulation . . . 15
3.5.3 Single-view fallback . . . 16
3.5.4 Wrist pose and gripper state . . . 16
3.6 External Hand-Joint Reference Protocol . . . 17
3.7 Object-Centric Representation . . . 17
3.8 Retargeting as a Single Cost Functional . . . 18
3.8.1 Formulation . . . 18
3.8.2 Confidence weighting . . . 18
3.8.3 Robustification . . . 19
3.8.4 Why a single functional . . . 19
3.8.5 Solution . . . 20
3.8.6 Interpolation . . . 20
3.9 Replay Validation . . . 20
3.10 Dataset Export . . . 21
3.11 Assumptions and Simplifications . . . 21

4 Implementation and Evaluation **23**

4.1 Software and Experimental Setup . . . 23
4.2 Capture and Hand Reconstruction . . . 23
4.3 External HO-Cap Hand-Reference Pilot . . . 24
4.4 Object Models, Retargeting and Physics Replay . . . 25
4.5 Local Layout Transfer and Learning . . . 26
4.6 Placement Quality and Contact Limitations . . . 27
4.7 Statistical Interpretation and Remaining Measurements . . . 27

5 Analysis and Discussion **29**
5.1 What the Results Establish . . . 29
5.2 Error Sources and Reference Frames . . . 29
5.3 Attribution and Statistical Validity . . . 30
5.4 Scope and Implications . . . 31

6 Conclusion **32**
6.1 Summary . . . 32
6.2 Answers to the Research Questions . . . 32
6.3 Claim Boundary and Next Validation . . . 33
6.4 Concluding Remarks . . . 33

References **34**

A Notation **39**

B Exported Dataset Schema **40**

C Solver Configuration **41**

List of Figures

3.1 Stages of the ViKi pipeline. Capture, perception and representation are embodiment-agnostic; retargeting, replay validation and export are specific to the target manipulator. The feedback edge indicates that episodes failing replay are either re-solved with modified weights or discarded. . . . 13

4.1 Episode inventory and pair-timing distribution. The recovered object bar includes independent scratch rebuilds; it is not an additional sequential stage after the original object bar. Histogram counts are paired frames, not independent recording trials. . . . 24

4.2 Matched-layout success for the learned correction and unchanged source commands; error bars show Wilson 95% intervals. The first test is conditioned on teacher-success selection; the second seed includes all 30 sampled layouts. Both use one source motion and one learned checkpoint. 27
4.3 Blue–green XY centre error before and after per-layout candidate selection in 18 already accepted simulator scenes. Points below the dashed identity line improved; the selection is in-sample and does not remove red support contact. 27

List of Tables

I Comparison of the Proposed System Against the Nearest Published Work Along the Six Criteria Defined in Section 2.7 10

I Capture and Hand-Stage Measurements for the 78-Episode Cohort 24
II HO-Cap Hand-Joint Pilot: Selected Pair Versus All Eight Views 25
III Object, Plan and Simulation Outcomes (Distinct Artifact Branches) 26
IV Paired MuJoCo Stack Outcomes for the Layout-Conditioned Policy 27

I Symbols Used Throughout the Thesis 39

I Per-Episode Dataset Fields 40

I Weights and Solver Parameters Used in the Reported Experiments 41

Chapter 1

Introduction

1.1 Background

Learning manipulation from demonstrations is limited by the cost of acquiring robot-executable examples. Teleoperation records robot states and actions directly, but needs a robot and a control interface during collection. Passive RGB-D capture instead records a person's unconstrained motion with cameras. It is cheaper to collect, but it does not itself provide robot actions: hand motion must be reconstructed, related to objects, and transferred to a robot with different geometry. Even a plausible joint trajectory need not lift or place an object when contacts and dynamics are simulated.

This thesis studies that chain as a source of training data. The target experiment is deliberately narrow: a person transfers a blue 40 mm cube from a red support cube to a green target cube; a simulated old-generation UR10 with a Robotiq 2F-85 gripper attempts the same task. The goal is not to claim a general-purpose manipulation policy, nor to claim deployment on a physical robot that is not available for this study.

1.2 Problem Statement

The central question is whether passive demonstrations provide a measurable advantage when learning a robot policy in physics simulation. Three distinct gaps intervene. Perception must recover consistent hand and object motion from two imperfectly calibrated RGB-D views. Retargeting must turn a human motion into a reachable robot plan with a meaningful gripper command. Finally, the plan must survive physical execution: a kinematically valid approach may still miss a cube, slip during transfer, or fail on release.

Object-relative trajectories offer a way to transfer demonstrations to new cube placements, but they require a reliable object pose and must be checked again after transfer. Synthetic variations are therefore candidates for training, not valid demonstrations merely because a geometric transform can be computed. A second distinction is between learning with object coordinates and learning from

images. The former isolates the quality of transferred data and control; the latter also tests perception and the visual simulation gap.

1.3 Research Gap

Prior work covers parts of this problem. Teleoperation research develops retargeting objectives, while passive-video systems estimate human and object motion. MimicGen demonstrates that object-centric subtask transfer plus simulation filtering can expand a small source dataset [1]. This work does not claim those operators as novel. Its question is whether an accessible, camera-only capture pipeline can produce useful robot-learning examples for the specified cube-transfer task, and which stage limits the yield. The comparison must include a controller that uses no ViKi examples; otherwise success in simulation cannot be attributed to the capture pipeline.

1.4 Research Questions and Hypotheses

RQ. *Can passive two-view RGB-D demonstrations processed by ViKi improve learning of a simulated UR10 cube-transfer policy, compared with the same training task without ViKi-derived examples?*

RQ1. What fraction of recorded episodes survives hand reconstruction, object tracking, retargeting and physical replay, and where are the dominant losses?

RQ2. Does object-aware, physics-validated transfer to new cube placements improve held-out task success over equal-size world-frame or hand-only training sets?

RQ3. Under matched observations, control, training budget and evaluation placements, do ViKi-derived examples improve policy success or sample efficiency relative to learning without those examples?

RQ4. Can a visual policy using simulated camera frames and robot state retain task success without privileged object coordinates at test time?

These are testable hypotheses, not findings. H1 predicts a nonzero yield of physically successful transferred episodes. H2 predicts that object-aware examples improve success on unseen, reachable cube placements when example count is controlled. H3 predicts that prior ViKi examples reduce the number of simulation interactions needed to reach a given success rate. H4 predicts that a camera-based policy trained from a successful state-based teacher can perform the task when evaluated using only deployment-available observations. Failure at any gate is a result to report, not a reason to relabel a geometric plan as a physically successful demonstration.

1.5 Contributions

The implemented contribution is an offline pipeline from calibrated RGB-D recordings to hand trajectories, object tracks, and UR10 retarget plans. The experimental contribution sought here is a reproducible assessment of processing yield, simulated physical validity, object-relative variation, and the marginal value of ViKi-derived data for state and visual policies. Simulation replay, learning outcomes and real-world transfer are not claimed until measured. The proposed experiment uses no motion-capture ground truth or physical UR10; geometric consistency metrics must therefore not be presented as absolute pose accuracy.

1.6 Thesis Structure

Chapter 2 reviews capture, retargeting and demonstration-driven robot learning. Chapter 3 defines the implemented pipeline and the planned simulation study. Chapter 4 separates current evidence from experiments still to run. Chapter 5 treats uncertainty and validity, and Chapter 6 states the claims that the eventual results can and cannot support.

Chapter 2

Literature Review

2.1 Preamble

This chapter surveys the state of knowledge relevant to converting passively recorded human manipulation demonstrations into robot-executable datasets. It is organised not chronologically but by the architectural elements of the proposed system, so that for every design decision taken in Chapter 3 the alternatives considered and the reason for the choice are visible.

Section 2.2 examines how the human hand is represented, from full parametric meshes to the minimal wrist-pose-plus-gripper abstraction. Section 2.3 contrasts sequential and single-cost-functional retargeting formulations and reviews the solvers used to realise them. Section 2.4 compares object-centric and world-frame task representations. Section 2.5 covers multi-camera RGB-D calibration, depth fusion and cross-device synchronisation, with emphasis on reported quantitative accuracy. Section 2.6 reviews dataset formats, downstream policies and demonstration curation. Section 2.7 positions ViKi against the nearest published systems, and Section 2.8 states the resulting methodological choices.

Sources were drawn from arXiv (`cs.RO`, `cs.CV`, `cs.LG`), the proceedings of CoRL, RSS, ICRA, IROS, CVPR and NeurIPS, peer-reviewed sensor and calibration literature, and vendor documentation for the depth cameras and solvers employed. Where a source is a recent preprint whose peer-review status could not be confirmed, this is stated explicitly.

2.2 Representing the Human Hand

2.2.1 Full parametric reconstruction

A large body of work recovers the complete kinematics of the demonstrator. OKAMI [11] combines SMPL-H for the body with MANO for the hand and retargets the two separately. DexCanvas [24] builds a human–robot correspondence dataset by fitting MANO to data from twenty-two infrared

motion capture cameras and two RGB-D sensors. DexMV [12] likewise recovers a full MANO hand together with object pose from video before translating the demonstration into simulation.

Full reconstruction yields a rich representation—the pose of every finger joint—and is necessary when the target embodiment is a multi-fingered dexterous hand. It also introduces a heavy perception stack and an additional, poorly characterised error source: the mesh fit itself. For a parallel-jaw gripper, most of that representational richness is discarded downstream.

2.2.2 Minimal representation: wrist pose and gripper state

The alternative, adopted in this work, represents the hand as a wrist pose in $\mathrm{SE}(3)$ together with a scalar or binary gripper state. This is not a concession made for tractability; it is the representation used by the mainstream of current teleoperation systems. Open-TeleVision [3] maps the operator's wrist pose to the arm end-effector through closed-loop inverse kinematics and encodes a parallel gripper by the single thumb–index vector. Bunny-VisionPro [4] and ACE [5] adopt the same decomposition, ACE additionally demonstrating that it transfers across dexterous hands, parallel grippers and quadrupeds.

Methodological support for down-weighting the wrist comes from the ablation study of Xin et al. [8], who assemble a comprehensive retargeting objective—global hand pose, overall hand shape, relative fingertip positions and fingertip orientations—and remove terms one at a time. They find that precise tracking of the global wrist pose is frequently unnecessary and can degrade fingertip accuracy, and consequently replace the wrist-position objective with a thumb-tip objective while regularising wrist orientation only weakly. The implication for a wrist-plus-gripper representation is that the wrist target should enter the optimisation as a weighted soft task rather than a hard constraint.

2.2.3 Egocentric and monocular capture

Several systems reduce capture-hardware requirements further. EgoMimic [17] records bare-handed demonstrations through Project Aria glasses and co-trains on human and robot data. DexWild [18] collects in-the-wild bare-hand data through a low-cost portable device and likewise co-trains. A recent egocentric RGB-D system [23] generates trajectories offline using analytic damped-least-squares inverse kinematics and replays them on a low-cost arm. These works confirm a trend toward lighter capture rigs, but each accepts monocular or single-view geometry, and none combines it with the metric multi-view accuracy that the present work requires.

2.3 Retargeting Formulations

2.3.1 The single cost functional

The prevailing modern formulation poses retargeting as a constrained nonlinear least-squares or quadratic program over the robot configuration $\boldsymbol{q}_t$ at each timestep, minimising a weighted sum of a data-fidelity term and a temporal smoothness term subject to joint limits. AnyTeleop [2] gives the canonical instance: the objective sums, over keypoints, the squared residual between a scaled human keypoint vector and the corresponding robot forward-kinematics vector, plus a first-order penalty on configuration change between consecutive frames. The accompanying `dex-retargeting` library exposes a vector optimiser for real-time teleoperation, a position optimiser explicitly recommended for offline dataset processing, and a contact-aware DexPilot optimiser; it is built on Pinocchio and solved with sequential quadratic programming.

Two refinements are directly relevant here. First, per-term confidence weighting: in the formulation reported by Open-TeleVision [3], each keypoint residual is scaled by a distance-dependent switching weight that increases when a pinch is detected. The mechanism is precisely analogous to weighting by a per-frame perception confidence, and it establishes that weighting individual data terms by a scalar signal is accepted practice. Second, second-order smoothness: DexFlow [6] augments the standard objective with a dynamic smoothing weight and Hessian regularisation to enforce C^2 continuity, penalising curvature and not merely velocity. Robustification by a Huber loss on the data term, reported in recent dexterous teleoperation work [7], provides a mechanism for rejecting perception outliers inside the optimiser rather than smoothing them away beforehand.

2.3.2 The sequential pattern and its critique

Against this, the sequential pattern—filter the observed task-space trajectory, solve inverse kinematics per frame, filter the joint output— remains common in offline video-to-robot pipelines. Its deficiency is structural rather than empirical. Filtering is applied in task space, whereas the smoothness requirement is on the joint trajectory; because inverse kinematics is nonlinear, a smooth Cartesian input does not imply a smooth joint output, particularly near singularities. Pre-filtering also attenuates genuine fast motion indiscriminately with noise. In the single-functional formulation the smoothness penalty acts directly on $\boldsymbol{q}_t$, so joint-space smoothness is obtained by construction.

2.3.3 Offline pipelines and their retargeting operators

ORION [10] does not perform per-frame hand retargeting: it constructs an object-centric plan over sparse keyframes from an open-world object graph. OKAMI [11] warps a reference trajectory to a new object position and then solves inverse kinematics—the sequential pattern. HOWTransfer [20], the closest published pipeline in workflow terms, performs passive markerless capture and transfers

trajectories using flow-matching $\mathrm{SE}(3)$ grasp sampling with Laplacian trajectory editing, not a single weighted cost functional; its representation is hand-centric rather than object-centric. No offline pipeline located in this review adopts the AnyTeleop-style single functional.

2.3.4 Solvers

Pinocchio underlies the retargeting stack of AnyTeleop, Open-TeleVision and ACE. Pink [9] exposes weighted inverse kinematics directly as a quadratic program: each task contributes a quadratic residual with a weight matrix normalising task coordinates to common units, while joint limits, velocity limits and self-collision barriers enter as inequality constraints, optionally with control-barrier formulations. This structure admits the entire proposed cost functional natively—a frame task for the wrist target, posture and velocity regularisation tasks for smoothness, and limits as constraints— solved as one program per frame. The solver choice made in Chapter 3 therefore lies on the mainstream path rather than diverging from it.

2.4 Task Representation

Object-centric representations dominate recent work on cross-environment and cross-embodiment generalisation. ORION [10] represents a task as the three-dimensional object trajectory relative to initial and final object positions, arguing that this generalises across visual backgrounds, camera shifts, spatial layouts and novel object instances, and that full-body reconstruction of the demonstrator is unnecessary. SPOT [13] encodes each task as the $\mathrm{SE}(3)$ object-pose trajectory relative to the target explicitly in order to decouple embodiment-specific actions from sensory input, enabling learning from action-free human demonstrations, and conditions a diffusion policy on the object trajectory. Flow-based formulations [14] make an equivalent argument while relaxing the rigid-object assumption that pose-trajectory methods require.

OKAMI [11] occupies an intermediate position: the representation is demonstrator-centric, but object-aware warping adapts trajectories to new object positions—an implicit acknowledgement that pure world-frame replay fails when the object moves. Vi-RoIL [21] is object-centric by construction, expressing wrist trajectories and keyframe hand poses in the object frame, but recovers metric scale from a monocular view of a marker cube and evaluates only in simulation.

The trade-off is that object-centric representations require robust object pose or segmentation, and pose-trajectory variants assume rigidity, whereas world-frame trajectories are simpler but tied to the recording layout. Storing both, as proposed in Section 3.7, defers the choice to the downstream consumer at negligible storage cost.

2.5 Multi-Camera RGB-D Calibration, Fusion and Synchronisation

HO-Cap provides eight calibrated RealSense views and semi-automatic MANO hand–object annotations [31]. It is an external reference for the hand-joint stage, not an optical-motion-capture measurement of our two-Kinect rig. Camera selection on the evaluated sequences must be reported as pilot selection rather than held-out performance.

2.5.1 Fiducial choice

ChArUco boards combine the sub-pixel corner accuracy of a checkerboard with the robust identification and occlusion tolerance of ArUco markers. A 6×8 board provides up to thirty-five corners per view against four for a single AprilTag, and remains detectable under partial occlusion and at the image boundary where a plain checkerboard fails. Reported reprojection errors for ChArUco and ArUco calibration are typically below half a pixel. AprilTags are easier to detect at long range and oblique incidence and suffice for extrinsic-only estimation, but are ill-suited to intrinsic calibration owing to the small number of corners. Current practice is therefore to use ChArUco for intrinsics and either fiducial for extrinsics, preferring AprilTags for oblique overhead views.

2.5.2 Reported accuracy of multi-Azure-Kinect rigs

Romeo et al. [25] provide the cleanest direct comparison of calibration procedures for multi-Kinect rigs, reporting that three-dimensional procedures outperform two-dimensional chessboard approaches, with mean distances between corresponding point-cloud points of 21.4 mm at colour resolution and 9.9 mm at infrared resolution for a static scene, and 20.9 mm and 7.4 mm respectively for a dynamic scene; body-joint alignment error was 35.4 mm. Darwish et al. [26] calibrate five Azure Kinects against a ChArUco-coded cube, combining reprojection error with point-to-plane and patch-to-plane terms to compensate depth noise. Kim et al. [27] perform coarse registration from body tracking followed by feature-match refinement, retaining above ninety-five per cent inliers. Bu and Lee [28] use markerless coarse calibration with coloured ICP refinement and report—importantly for rig geometry—that ICP fails for near-opposing camera placements owing to insufficient point-cloud overlap.

Single-sensor specifications bound the achievable accuracy from below. The Azure Kinect DK is specified at a random error standard deviation of at most 17 mm and a typical systematic error below 11 mm plus 0.1 per cent of distance [32]; independent evaluation confirms these figures [29], and comparative work finds improved spatial and temporal accuracy relative to Kinect v2 beyond 2.5 m [30]. The Intel RealSense D435i is specified at depth accuracy below one per cent of distance [33]. The practical consequence is that centimetre-level, not millimetre-level, agreement should be

expected between calibrated views.

2.5.3 Depth fusion

Naive averaging of overlapping depth observations is inferior to confidence-weighted fusion. Weighting by distance and by the angle between the surface normal and the viewing direction—so that surfaces observed near perpendicular contribute more than obliquely observed ones—is established practice [36]. Learned confidence maps estimated from the depth signal itself are used to re-weight depth features before fusion [37], and surfel-based reconstruction attaches an explicit confidence value to each element and performs curvature-weighted registration [38]. Both sensor families used in this work expose per-pixel invalidation or confidence signals that can drive such weights directly.

2.5.4 Cross-device synchronisation

Azure Kinect devices support hardware synchronisation through daisy-chained or star-wired trigger cabling, with subordinate devices required to start before the master and depth exposures staggered to prevent laser interference; the vendor documentation states explicitly that equal timestamps between depth and colour streams, or between cameras, are not guaranteed [32]. RealSense devices provide an analogous inter-camera synchronisation mode, with the counter-intuitive property that timestamps appear aligned when hardware synchronisation is disabled and exhibit bounded drift when it is enabled, the drift itself serving as verification [33, 34].

The pitfall for a heterogeneous rig is that hardware clocks on different device families are not mutually comparable, since each counts its own cycles from its own epoch. Documented practice for such rigs is to stamp every arriving frame with a single host-side monotonic clock and to align streams by nearest timestamp in that common domain, accepting millisecond-scale precision [34]. Because Azure Kinect and RealSense triggers are mutually incompatible, this is the only available mechanism for aligning the two families, and it is the mechanism adopted in Section 3.4.

2.5.5 Hand–eye calibration

The OpenCV `calibrateHandEye` routine implements the Tsai–Lenz, Park–Martin, Horaud–Dornaika, Andreff and Daniilidis solvers. A comparative noise study [35] finds the Daniilidis, Chou and Park methods most robust to translation noise, with Daniilidis yielding the best translation estimates, while Tsai, Li and Lu degrade as rotation noise grows. Methods that decouple rotation from translation are faster but propagate rotation error into the translation estimate; Daniilidis solves both simultaneously through dual quaternions. Because the present system anchors cameras and robot to a shared ChArUco workspace frame, classical hand–eye estimation is largely bypassed; where an explicit camera-to-robot transform is required, Daniilidis is used.

2.6 Dataset Generation and Downstream Policies

The LeRobot HDF5 conventions together with ACT and Diffusion Policy constitute the default imitation-learning toolchain and define the export target of this work. OpenVLA-OFT [16] reports that parallel decoding, action chunking, a continuous action representation and $L1$ regression raise average success across four task suites from 76.5 to 97.1 per cent while increasing action-generation throughput twenty-six-fold, and is therefore a credible alternative consumer of the generated data.

Curation matters as much as generation. Demo-SCORE [15] trains a classifier on policy rollouts to filter low-quality demonstrations and reports an absolute success-rate improvement exceeding fifteen to thirty-five per cent over training on all demonstrations. Smoothness-based scoring of trajectories has been reported to yield comparable gains at a fraction of the data **[citation to be confirmed]**. This motivates treating dataset curation as an explicit pipeline stage rather than an afterthought, and—in the present design—moving it upstream of export in the form of physical replay screening.

2.7 Positioning of the Proposed System

Table I compares ViKi against the nearest published systems along the six criteria that jointly define the proposed architecture: (C1) passive markerless capture without teleoperation hardware; (C2) multi-camera metrically calibrated RGB-D sensing; (C3) object-centric dense trajectory representation; (C4) retargeting as a single confidence-weighted cost functional; (C5) export of an imitation-learning-ready dataset; and (C6) physical replay used to validate the dataset before export.

TABLE I
Comparison of the Proposed System Against the Nearest Published Work Along the Six Criteria Defined in Section 2.7

System	C1	C2	C3	C4	C5	C6
ViKi (proposed)	✓	✓	✓	✓	✓	✓
HOWTransfer [20]	✓	∼	—	—	∼	✓
Vi-RoIL [21]	∼	—	✓	—	∼	—
ORION [10]	✓	∼	✓	—	—	—
OKAMI [11]	✓	∼	∼	—	∼	—
EgoMimic [17]	✓	—	—	—	∼	—
DexWild [18]	∼	—	—	—	∼	—
Tool-as-Interface [19]	✓	∼	∼	—	∼	—
RoboPaint [22]	—	✓	∼	—	✓	✓
DexCanvas [24]	—	✓	—	—	∼	—

✓ = satisfied; ∼ = partially satisfied; — = not satisfied.

No located system satisfies all six criteria simultaneously. HOWTransfer [20] is closest in workflow, satisfying C1 and C6, but is hand-centric rather than object-centric, uses calibrated stereo RGB rather than metric RGB-D, and does not employ a single cost functional. Vi-RoIL [21] is closest in representation, satisfying C3, but relies on a monocular view of a marker cube and is evaluated only

in simulation. The intersection of the online joint-optimisation retargeting literature with the offline object-centric dataset-generation literature is therefore unoccupied, and it is that intersection which this work targets.

Several of the systems listed are recent preprints [20, 22, 23, 24] whose peer-review status could not be confirmed at the time of writing; their reported results are treated as provisional.

2.8 Conclusion of the Review

The review supports the following methodological choices, which Chapter 3 implements.

1. *Hand representation.* A wrist pose in $\mathrm{SE}(3)$ with a binary gripper state is adopted, following [3, 4, 5], and the wrist target enters as a weighted soft task following the ablation evidence of [8]. Parametric mesh reconstruction is rejected as unnecessary for a parallel-jaw target and as an additional error source.

2. *Retargeting.* A single weighted cost functional is adopted, following [2, 6], with confidence weighting after [3] and Huber robustification after [7]. The sequential smooth–IK–smooth pattern is rejected on the structural grounds set out in Section 2.3.

3. *Solver.* Pink on Pinocchio [9] is adopted, since it expresses the entire functional natively as a constrained quadratic program and is the solver family used by the reference teleoperation systems.

4. *Task representation.* Object-relative trajectories are adopted following [10, 13], and stored alongside workspace-anchored trajectories so that the choice is deferred downstream.

5. *Sensing.* ChArUco workspace anchoring is adopted; centimetre-level cross-camera agreement is assumed on the evidence of [25]; camera placement preserves substantial view overlap on the evidence of [28]; depth is fused with distance, incidence-angle and sensor confidence weights following [36, 37].

6. *Synchronisation.* The two Azure Kinects are hardware synchronised; the RealSense, which cannot share their trigger, is aligned through a single host-side monotonic clock following [34].

7. *Curation.* Dataset curation is treated as an explicit stage and moved upstream of export in the form of physical replay screening, motivated by [15] and by the replay-based validation of [20].

Chapter 3

Design and Methodology

3.1 System Overview

ViKi is an offline pipeline: demonstrations are recorded, processed and exported in distinct stages, with no requirement that any stage run at interactive rates. This is a deliberate design choice. Relaxing the real-time constraint that shapes teleoperation systems permits global optimisation over whole trajectories, iterative refinement, and a validation stage that executes on physical hardware—none of which is available to an online system.

The pipeline comprises six stages, shown in Fig. 3.1.

1. *Capture.* Synchronised RGB-D streams are recorded from a calibrated multi-camera rig while a demonstrator performs the task bare-handed.

2. *Perception.* Hand landmarks are detected per view, lifted to three dimensions using sensor depth, and fused across views with confidence weights to yield a wrist pose in $\mathrm{SE}(3)$ and a gripper state.

3. *Representation.* The wrist trajectory is expressed both relative to the manipulated object and relative to the ChArUco workspace anchor.

4. *Retargeting.* A robot joint trajectory is obtained by minimising a single weighted cost functional subject to kinematic and collision constraints.

5. *Replay validation.* The synthesised trajectory is executed on the physical manipulator; proprioception is recorded and infeasible episodes are rejected.

6. *Export.* Surviving episodes are written in a LeRobot-compatible format for downstream policy training.

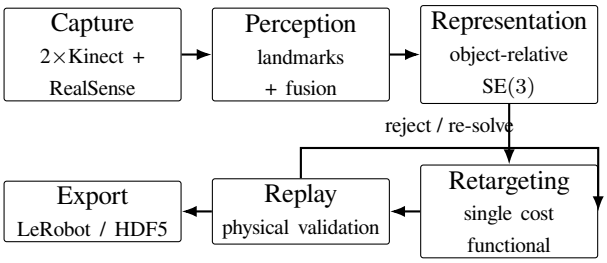


Fig. 3.1. Stages of the ViKi pipeline. Capture, perception and representation are embodiment-agnostic; retargeting, replay validation and export are specific to the target manipulator. The feedback edge indicates that episodes failing replay are either re-solved with modified weights or discarded.

3.2 Hardware and Workspace

The capture rig comprises two Azure Kinect DK cameras and one Intel RealSense D435i observing a tabletop workspace, with a UR10 manipulator mounted at a fixed position relative to that workspace. Reach is the binding constraint on this choice rather than payload: with the base offset used here, target poses lie between 525 and 911 mm from the base, so a UR3, whose reach is 500 mm, cannot cover the demonstrated workspace at all—a substitution verified to leave between 12 and 61 per cent of every recording unreachable, with tracking error on those frames two to three times that on reachable ones. The Azure Kinect was selected for the primary views because its time-of-flight depth sensing provides substantially better range estimation than neural regression from RGB, and because its specified error characteristics—random error standard deviation at most 17 mm and systematic error below 11 mm plus 0.1 per cent of distance [32]—are documented and independently verified [29]. The RealSense provides a supplementary close-range view at higher depth accuracy [33].

The two Kinects are placed to preserve substantial overlap in their fields of view. This is a direct consequence of the finding of Bu and Lee [28] that iterative-closest-point refinement fails between near-opposing views for lack of point-cloud overlap; near-orthogonal placement retains both triangulation baseline and overlap.

3.3 Calibration

3.3.1 Intrinsics and extrinsics

Intrinsic parameters for every colour and depth stream are estimated from ChArUco board observations. Extrinsic calibration follows the workspace-anchor principle: rather than expressing cameras relative to one another, each camera is calibrated against a ChArUco board rigidly fixed to the workspace, defining a common world frame W. For camera k this yields

$$\boldsymbol{T}_{C_k}^{W} \in \mathrm{SE}(3), \tag{3.1}$$

the transform from the camera frame C_k to the workspace frame W. Anchoring to the workspace rather than to a reference camera has two consequences: adding or moving a camera does not invalidate the calibration of the others, and the robot base can be registered into the same frame without a separate camera-to-camera chain.

3.3.2 Robot registration

The manipulator base frame R is registered to W by observing a ChArUco target held in a known end-effector pose across a set of configurations, giving $\boldsymbol{T}_R^W$ directly. Where an explicit camera-to-robot transform is required instead, the Daniilidis dual-quaternion solver is used, following the robustness evidence of [35], with fifteen to twenty poses spanning a hemisphere.

3.4 Synchronisation

The two Azure Kinects are hardware-synchronised in master/subordinate configuration, with subordinate devices started before the master and depth exposures staggered to avoid mutual laser interference [32]. The RealSense cannot share that trigger; its synchronisation mode is incompatible at the electrical level.

Cross-family alignment therefore uses a single host-side monotonic clock. Every frame, from every device, is stamped on arrival with

$$\tau = \texttt{time.monotonic_ns()}, \tag{3.2}$$

and streams are aligned by nearest neighbour in τ. Two points deserve emphasis. First, per-device hardware timestamps are *not* used for cross-device arithmetic: distinct device families count their own cycles from their own epochs, and the vendor documentation states explicitly that equal timestamps across cameras are not guaranteed [32]. Second, a monotonic clock rather than wall-clock time is required, since wall-clock time may be adjusted discontinuously by time synchronisation daemons, which would corrupt interval computations mid-recording.

Residual alignment jitter is measured rather than assumed (Section 4.2), and hardware synchronisation of the Kinect pair is verified by confirming bounded drift in the inter-camera timestamp difference [34].

3.5 Perception and Multi-View Reconstruction

3.5.1 Landmark detection and lifting

Hand landmarks are detected in each colour stream, yielding for view k at time t a set of image-plane landmarks with associated visibility scores $v_{i,t}^k \in [0,1]$. Each landmark is lifted to three dimensions using the registered depth map rather than a learned depth estimate, exploiting the time-of-flight measurement directly, and transformed into the workspace frame by (3.1).

3.5.2 Multi-view triangulation

When a landmark is seen by two or more calibrated views, its position is not averaged from per-view depth lifts but estimated directly from the image observations, with depth entering as an additional soft residual rather than as the anchor. Averaging per-view lifts inherits the depth error of every view; triangulation is constrained by the camera geometry, which is known to calibration accuracy and does not degrade with range.

For landmark i at time t, let $\mathcal{K}_{i,t}$ be the views whose detector score clears a threshold and whose observation is finite. Every pair $(a,b) \subset \mathcal{K}_{i,t}$ yields a linear direct-linear-transform estimate, which is retained only if it lies in front of both cameras and if the two viewing rays subtend at least a minimum angle θ_{min}; a pair with a short effective baseline is ill-conditioned regardless of how confident the detector is. Each surviving hypothesis is scored against all views in $\mathcal{K}_{i,t}$ by the number of reprojection inliers, and the best is refined over its inlier set by

$$\hat{\boldsymbol{x}}_{i,t} \;=\; \arg\min_{\boldsymbol{x}} \sum_{k \in \mathcal{K}_{i,t}^{\text{in}}} \rho\Big(\big\| \pi_k(\boldsymbol{x}) - \boldsymbol{u}_{i,t}^k \big\|^2 \Big) \;+\; \lambda_d \sum_k \beta_{i,t}^k \, \rho_d\Big(\big(z_k(\boldsymbol{x}) - d_{i,t}^k - \delta_i \big)^2 \Big), \tag{3.3}$$

where π_k is the pinhole projection of view k, $\boldsymbol{u}_{i,t}^k$ the observed pixel, z_k the depth of $\boldsymbol{x}$ in that camera, $d_{i,t}^k$ the measured depth, and δ_i a per-landmark offset from the skin surface to the joint centre—depth sees the skin, the joint sits inside the finger. The depth weight $\beta_{i,t}^k$ is attenuated by the local depth spread and set to zero for invalid samples, so that a single bad silhouette pixel cannot move an otherwise well-constrained point. Observations are undistorted once, before the linear estimate, and every projection downstream is a plain pinhole map; distortion is never re-applied inside the residual.

The estimate carries a transparent geometric quality

$$q_{i,t} \;=\; \underbrace{\frac{|\mathcal{K}_{i,t}^{\text{in}}|}{|\mathcal{K}_{i,t}|}}_{\text{agreement}} \cdot \underbrace{\text{clip}\Big(1 - \frac{\bar{e}_{i,t}}{2 e_{\text{in}}},\, 0,\, 1 \Big)}_{\text{reprojection}} \cdot \underbrace{\text{clip}\Big(\frac{\theta_{i,t}}{\theta_{\text{ref}}},\, 0,\, 1 \Big)}_{\text{baseline}}, \tag{3.4}$$

with $\bar{e}_{i,t}$ the mean reprojection error over inliers, $\theta_{i,t}$ the widest ray angle, and e_{in} the inlier radius. This is a transparent heuristic, not a calibrated variance, and it is used as such: it weights evidence,

and is never interpreted as an inverse covariance.

A property of (3.4) deserves explicit statement, because it governs how much of a recording survives. The same e_{in} is the acceptance threshold, the scale of the robust loss ρ in (3.3), and the denominator of the reprojection factor. The factor is therefore designed to grade confidence out to $2e_{\text{in}}$, while the threshold admits nothing beyond e_{in}: half of the designed grading range is unreachable by construction. Widening e_{in} does not introduce a soft-weighting mechanism, it permits the existing one to operate over its own range. A consequence is that q is not comparable across settings of e_{in}: the same residual scores differently under a different denominator.

3.5.3 Single-view fallback

Where only one view observes a landmark, triangulation is undefined and the estimate falls back to the depth lift of that view. The historical multi-view path also combined per-view lifts by confidence-weighted averaging,

$$\hat{\boldsymbol{x}}_{i,t} \;=\; \frac{\sum_k w_{i,t}^k \, \boldsymbol{x}_{i,t}^k}{\sum_k w_{i,t}^k}, \qquad w_{i,t}^k \;=\; \underbrace{v_{i,t}^k}_{\text{detector}} \cdot \underbrace{c_{i,t}^k}_{\text{sensor}} \cdot \underbrace{\frac{1}{\left(d_{i,t}^k\right)^2}}_{\text{range}} \cdot \underbrace{\max\!\left(0, \cos\theta_{i,t}^k\right)}_{\text{incidence}}, \tag{3.5}$$

where $v_{i,t}^k$ is the detector score, $c_{i,t}^k$ the sensor depth validity at that pixel, $d_{i,t}^k$ the range, and $\theta_{i,t}^k$ the incidence angle. The range and incidence factors follow established practice in geometric depth fusion, where surfaces observed near perpendicular are weighted above obliquely observed ones because they are measured more accurately [36]; the detector and sensor factors follow confidence-map re-weighting [37]. The product form ensures that a zero in any factor—an undetected landmark, an invalid depth pixel, a grazing view— removes that view's contribution entirely rather than attenuating it.

This path is retained for reproducibility of earlier recordings and for single-camera operation. It is not the path used for the results reported here, because averaging per-view lifts propagates depth error into the fused point, whereas (3.3) constrains the point by geometry and admits depth only as evidence.

3.5.4 Wrist pose and gripper state

The wrist pose $\boldsymbol{T}_{H,t}^{W} \in \mathrm{SE}(3)$ is obtained by fitting an orthonormal frame to the fused wrist and knuckle landmarks. The gripper state is the thumb–index separation, mapped to a normalised opening,

$$g_t \;=\; \mathrm{clip}\!\left(\frac{\left\| \hat{\boldsymbol{x}}_{\text{thumb},t} - \hat{\boldsymbol{x}}_{\text{index},t} \right\| - \tau_{\text{closed}}}{\tau_{\text{open}} - \tau_{\text{closed}}},\, 0,\, 1 \right), \tag{3.6}$$

temporally filtered to suppress chatter. An earlier formulation thresholded the same distance into a binary open/closed indicator; it was replaced because a parallel-jaw gripper is a continuously com-

mandable actuator, and quantising the one degree of freedom that carries the grasp discards precisely the approach and release phases that distinguish a successful demonstration from a failed one. This single scalar remains the entire hand representation beyond the wrist frame, following the convention established for parallel-jaw targets in [3, 5].

3.6 External Hand-Joint Reference Protocol

Two right-hand HO-Cap subject-1 sequences are imported with 120 frames from each of eight RealSense cameras. They are separate from the local cube cohort and do not measure its Azure Kinect calibration. Published camera poses and MANO hand-joint annotations are expressed in one world frame; Extract and Prepare run without access to the reference labels. Predicted 21-joint 3-D positions are matched by exact imported frame index and synthetic capture timestamp, with no root alignment, Procrustes fit, time search or reference-guided frame filtering. MPJPE is the mean Euclidean distance over finite matched joints; coverage is matched joints divided by all valid annotated joints. P95 and PCK@20 mm are reported alongside coverage. Partially written `cln.npz` files are excluded unless Prepare marks the stage complete. The two-view pair was selected after screening all 28 pairs on these same sequences, so its scores are in-sample pilot results.

3.7 Object-Centric Representation

Let $\boldsymbol{T}_{O,t}^{W}$ denote the pose of the manipulated object in the workspace frame. Each demonstration is stored in two forms. The workspace-anchored form is $\boldsymbol{T}_{H,t}^{W}$ directly; the object-relative form is

$$\boldsymbol{T}_{H,t}^{O} \;=\; \left(\boldsymbol{T}_{O,t}^{W}\right)^{-1} \boldsymbol{T}_{H,t}^{W}. \tag{3.7}$$

The object-relative form is what transfers. If the same task is to be performed with the object at a new pose $\boldsymbol{T}_{O'}^{R}$ expressed in the robot frame, the desired end-effector pose at time t is

$$\boldsymbol{T}_{E,t}^{R,\mathrm{des}} \;=\; \boldsymbol{T}_{O'}^{R}\, \boldsymbol{T}_{H,t}^{O}\, \boldsymbol{T}_{E}^{H}, \tag{3.8}$$

where $\boldsymbol{T}_{E}^{H}$ is a fixed offset relating the human wrist frame to the robot end-effector frame, determined once per embodiment. Trajectories are retained densely rather than reduced to keyframes, since the continuous velocity profile of the demonstration carries task-relevant information— approach speed, contact timing—that sparse keyframes discard.

Both representations are exported. Storing both costs little and defers to the downstream consumer a choice that the literature does not settle universally: object-relative trajectories generalise across object placement [10, 13] but depend on object pose estimation, whereas workspace-anchored trajectories are simpler and adequate within a fixed layout.

3.8 Retargeting as a Single Cost Functional

3.8.1 Formulation

Let $\boldsymbol{q}_t \in \mathbb{R}^n$ be the manipulator configuration at time t and $f(\boldsymbol{q}_t) \in \mathrm{SE}(3)$ the forward kinematics of the end-effector. The task-space residual with respect to the desired pose (3.8) is the logarithmic map of the pose error,

$$\boldsymbol{e}_t(\boldsymbol{q}_t) \;=\; \mathrm{Log}\!\left(\left(\boldsymbol{T}_{E,t}^{R,\mathrm{des}}\right)^{-1} f(\boldsymbol{q}_t)\right)^{\vee} \in \mathbb{R}^6. \tag{3.9}$$

Retargeting is then posed as the single constrained problem

$$\begin{aligned}\min_{\boldsymbol{q}_{1:T}} \;\; J(\boldsymbol{q}_{1:T}) \;=\; \sum_{t=1}^{T} \Big[& \underbrace{\omega_t\, \rho_\delta(\|\boldsymbol{e}_t(\boldsymbol{q}_t)\|_{\boldsymbol{\Sigma}})}_{\text{data fidelity}} \;+\; \underbrace{\lambda_v \left\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\right\|^2}_{\text{velocity}} \\ & +\; \underbrace{\lambda_a \left\|\boldsymbol{q}_t - 2\boldsymbol{q}_{t-1} + \boldsymbol{q}_{t-2}\right\|^2}_{\text{acceleration}} \;+\; \underbrace{\lambda_r \left\|\boldsymbol{q}_t - \boldsymbol{q}_{\mathrm{ref}}\right\|^2}_{\text{posture}} \Big]\end{aligned} \tag{3.10}$$

subject to

$$\boldsymbol{q}_{\min} \;\le\; \boldsymbol{q}_t \;\le\; \boldsymbol{q}_{\max}, \tag{3.11}$$

$$\left\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\right\| \;\le\; \dot{\boldsymbol{q}}_{\max}\, \Delta t, \tag{3.12}$$

$$h_j(\boldsymbol{q}_t) \;\ge\; 0, \quad j = 1, \ldots, m, \tag{3.13}$$

where

ω_t	per-frame perception confidence weight, defined in (3.14);
ρ_δ	Huber loss with transition parameter δ;
$\boldsymbol{\Sigma}$	diagonal matrix normalising translational and rotational residual units;
$\lambda_v, \lambda_a, \lambda_r$	weights on velocity, acceleration and posture regularisation;
h_j	collision and self-collision barrier functions.

3.8.2 Confidence weighting

The per-frame weight aggregates the geometric qualities (3.4) of the landmarks that define the palm frame at time t,

$$\omega_t \;=\; \left(\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} c_{i,t}\right)^{\alpha} \cdot \mathbb{1}[\text{frame valid}]\,, \qquad c_{i,t} = \begin{cases} q_{i,t}, & \text{landmark triangulated,} \\ 0, & \text{landmark interpolated,} \end{cases} \tag{3.14}$$

with $\mathcal{I}$ the four landmarks defining the palm frame—wrist and the index, middle and little metacarpophalangeal joints—and α controlling how sharply low-confidence frames are discounted.

Equation (3.14) as written admits a frame whose palm was only partially observed: a mean over four terms is non-zero whenever one of them is, so a palm with a single triangulated landmark still contributes at roughly a quarter weight, and the orientation it contributes was fitted partly through interpolated points. Measured across five recordings, between 45 and 61 per cent of the frames carrying non-zero weight had an incompletely observed palm. A stricter variant zeroes the frame unless every landmark in $\mathcal{I}$ was triangulated,

$$\omega_t^{\text{strict}} \;=\; \omega_t \cdot \mathbb{1}\big[\,\forall i \in \mathcal{I} :\; i \text{ triangulated}\,\big]\,. \tag{3.15}$$

Both are reported, because the stricter form is not uniformly better: it removes a fabrication, but it also removes between 45 and 61 per cent of the weighted frames, and the measurements in Chapter 5 show the solver drifting further from the target for want of the support the fabricated frames were supplying. Recovering that support—rather than reinstating the fabrication—is identified as future work in Section 5.4. The construction is the offline analogue of the distance-dependent switching weights used in online teleoperation retargeting [3]: frames in which the hand is partially occluded or poorly observed exert proportionally less influence on the solution, and the smoothness terms carry the trajectory through them instead of the data term dragging it toward a spurious observation.

3.8.3 Robustification

The Huber loss ρ_δ bounds the influence of gross outliers—a mis-detected hand, a depth dropout that survives the confidence weighting—at the level of the data term itself, rather than allowing a filtering stage to smear the outlier across neighbouring frames [7]. This is the mechanism that replaces pre-smoothing: outliers are down-weighted where they enter, not attenuated after they have already propagated.

3.8.4 Why a single functional

Equation (3.10) differs from the sequential alternative in one structurally important respect. The smoothness penalties act on $\boldsymbol{q}_t$, in the space where smoothness is required, and they are traded against data fidelity by the solver rather than applied blindly beforehand. In the sequential pipeline, task-space filtering cannot guarantee joint-space smoothness because the forward kinematics is nonlinear, and it cannot distinguish task-relevant fast motion from noise because it operates without reference to the data term. Here the trade-off is explicit: on high-confidence frames ω_t is large and the solution tracks the observation; on low-confidence frames the smoothness terms dominate and the solution interpolates.

Second-order smoothness, expressed by the acceleration term, follows the C^2-continuity argument

of DexFlow [6]. A trajectory that is first-order smooth may still exhibit curvature discontinuities that appear as jerk on the physical manipulator and as label noise to a policy trained on the data.

3.8.5 Solution

The problem is solved with Pink on Pinocchio [9]. Each term of (3.10) maps onto a weighted task in the quadratic program—a frame task for the data term, posture and velocity regularisation tasks for the smoothness terms—while (3.11)–(3.13) enter as inequality constraints, with collision avoidance expressed through control barrier functions. Levenberg–Marquardt damping stabilises the solution near singularities.

Because the setting is offline, the problem is solved over the whole trajectory rather than frame by frame, so the acceleration term is well defined at every interior timestep and the solution is not constrained by the causality requirement that an online solver faces. The trajectory is initialised from the previous solution warm-started along t, and boundary conditions at $t = 1, 2$ are handled by replicating the initial configuration.

3.8.6 Interpolation

Where gaps remain after fusion—landmarks lost across all views—the trajectory is completed by cubic spline interpolation over $\mathrm{SE}(3)$ prior to optimisation. Dynamic filters that impose a motion model, such as Kalman or extended Kalman filters, are deliberately not applied: they introduce a prior about the dynamics that is subsequently re-estimated by the optimiser, and the interaction between the two priors is difficult to characterise. The optimiser already imposes the required regularity through λ_v and λ_a.

3.9 Replay Validation

Synthesised trajectories are executed on the physical manipulator before the dataset is finalised. The stage serves three purposes.

First, it filters infeasibility. A trajectory may satisfy (3.11)–(3.13) in the kinematic model and still fail on hardware through controller tracking limits, unmodelled collisions or workspace boundary conditions. Such episodes are rejected rather than exported.

Second, it closes the embodiment gap at the source. The proprioceptive states recorded during replay are those the robot actually attained, not those the optimiser predicted. Exporting the recorded rather than the synthesised states means the dataset contains honest observation–action pairs from the target embodiment, which is precisely the property that teleoperated datasets possess by construction and that naive video-derived datasets lack.

Third, it provides a curation signal. Episodes are ranked by tracking residual during replay, and

the ranking is available to downstream consumers as a quality score, in the spirit of rollout-based demonstration curation [15] but applied before rather than after policy training.

Failure of an episode triggers either rejection or re-solution of (3.10) with modified weights—typically an increased λ_v or a relaxed data weight in the region of failure—as indicated by the feedback edge in Fig. 3.1. The number of re-solution attempts is bounded to prevent overfitting the trajectory to the replay controller.

3.10 Dataset Export

Surviving episodes are written in a LeRobot-compatible HDF5 layout. Each episode contains the synchronised camera streams, the recorded joint states and gripper states from replay, the object-relative and workspace-anchored wrist trajectories, per-frame confidence weights, and the replay tracking residual. Retaining the confidence weights and residuals in the exported artefact allows downstream curation to be repeated with different thresholds without re-recording.

3.11 Assumptions and Simplifications

The following assumptions delimit the scope of the work and are revisited in Section 5.4.

1. *Parallel-jaw target.* The hand is reduced to a wrist pose and a single continuous gripper opening (3.6). Multi-fingered dexterous targets would require the finger-level representation rejected in Section 2.2.

2. *The gripper body is not the hand.* The target is the midpoint between thumb and index tips, but a parallel-jaw gripper interposes a body of its own between the flange and that point, and the whole assembly must stay above the workspace plane. For tasks performed against the table surface the demonstrated pose can therefore be kinematically unreachable even when the arm's reach is ample: in one recording the target sits a median of 36 mm above the plane while the achieved tool point is held 87–145 mm above it, and the resulting vertical gap accounts for roughly 70 per cent of the squared tracking error. Improving perception makes this discrepancy more visible, not less, because interpolation had been smoothing the lowest excursions upward.

3. *Rigid objects.* The object-relative representation (3.7) presupposes a rigid body with a well-defined pose. Deformable objects would require a flow-based representation [14].

4. *Single-arm, quasi-static tasks.* The task class is restricted to single-arm tabletop manipulation at moderate speed, where the quasi-static assumption underlying purely kinematic retargeting holds.

5. *Fixed workspace.* The ChArUco anchor is rigidly attached to the workspace and the camera rig is not moved within a recording session.

6. *Kinematic-only transfer.* Forces and contact dynamics are not estimated. Tasks whose success depends on force regulation are outside the scope.

Chapter 4

Implementation and Evaluation

4.1 Software and Experimental Setup

The Python implementation separates capture, calibration, perception, preparation, retargeting and export. FastAPI serves offline jobs and the browser inspection interface; MuJoCo experiments use a separate image. Each stage writes a durable episode artifact. The `rec.npz` hand extraction, `cln.npz` prepared motion, `object_models.npz` scene model and `plan.h5` robot plan are not interchangeable evidence. Physical-robot replay remains a stub and is not used here.

One demonstrator moved a blue 40 mm cube from a red support cube to a green target cube on a 1200×700 mm table. The nominal red–green separation is 400 mm. Two hardware-synchronized Azure Kinects, each on an independent USB host controller, recorded the demonstrations. The robot is a simulated old-generation UR10 with a Robotiq 2F-85 two-finger gripper; no physical UR10 was present. The analysis cohort is the 78 nonempty episodes listed in `data/experiments/2026-10-09-full-audit.json`; it is not all directories or failed capture attempts. The source episodes share one approximately fixed cube arrangement. Local-cohort numbers were computed from the then-saved artifacts by `scripts/build_thesis_results.py`. Those local source artifacts were subsequently lost; the numerical record must be regenerated from a new cohort before a reproducible submission.

4.2 Capture and Hand Reconstruction

The 78 episodes contain 11,688 paired frame groups with device timestamps for both cameras. The absolute device-timestamp gap has a median of $155\ \mu$s, a 95th percentile of $167\ \mu$s and a maximum of $178\ \mu$s; none exceeds 1 ms. This is a pair-timing diagnostic, not a bound on RGB–depth exposure differences or 3-D position error. The host-arrival `sync_bounded` flag measures another clock and must not be substituted for the device gap. The large 400×500 mm ChArUco calibration made cloud overlap visually usable after an earlier small-board problem, but no independent measurement

certifies millimetre accuracy.

The multi-view hand stage solved 232,368 of 244,671 joint slots (95.0%) across the 78 episodes. Episode-wise coverage has a median of 97.6% and 10th–90th percentiles of 88.9–100.0%. Seventy-three episodes passed the articulated Prepare gate; five retain only partial `cln.npz` checkpoints and are *not* valid prepared demonstrations. In the 73 completed artifacts, 10,463 of 10,993 frames (95.2%) carry supported geometry. The longest unsupported run per episode has median 3 frames, 95th percentile 21.2 and maximum 53. These are coverage measurements, not absolute hand-pose errors for the local Kinect rig: there is no independent motion-capture reference. The separate HO-Cap experiment below measures a different capture system against its published annotations.

TABLE I
Capture and Hand-Stage Measurements for the 78-Episode Cohort

Quantity	Measured value
Paired two-Kinect frame groups	11,688 in 78 episodes
Device-timestamp pair gap, median / p95 / max	155 / 167 / 178 μs
Triangulated hand-joint slots	232,368 / 244,671 (95.0%)
Episodes passing articulated Prepare gate	73 / 78 (93.6%)
Supported frames among passed episodes	10,463 / 10,993 (95.2%)
Longest unsupported run, median / p95 / max	3 / 21.2 / 53 frames

Historical figure unavailable after local workspace loss. Regenerate from the source episode artifacts before submission.

Fig. 4.1. Episode inventory and pair-timing distribution. The recovered object bar includes independent scratch rebuilds; it is not an additional sequential stage after the original object bar. Histogram counts are paired frames, not independent recording trials.

4.3 External HO-Cap Hand-Reference Pilot

Two subject-1 sequences (120 frames each) were imported from HO-Cap, not from the local Kinect cube cohort. Screening the 28 camera pairs on these same frames selected `043422252387` and `105322251225`; all eight views were also imported and processed with the same Extract and Prepare stages. Table II compares absolute 21-joint world-frame predictions with the published semi-automatic MANO annotations using Section 3.6. Each row matched all 2,520 annotated joint slots. The numbers were recovered from the original run log after the local episode artifacts were lost, so this is a historical pilot, not a currently reproducible or held-out result.

Adding six cameras reduced MPJPE by 1.17 mm on `170105` but changed it by only 0.004 mm on `170650`. One view returned no hand observations in `170105`; fusion remained robust to it on this bounded sample. The selected pair still fails one colour–depth consistency check. These annotations do not ground-truth the local Kinect calibration, object pose, retargeting or robot execution.

TABLE II
HO-Cap Hand-Joint Pilot: Selected Pair Versus All Eight Views

Sequence	Views	Coverage	MPJPE (mm)	P95 (mm)	PCK@20 (%)
`170650`	selected pair	2,520/2,520	11.33	22.71	87.70
`170650`	all eight	2,520/2,520	11.34	22.17	87.50
`170105`	selected pair	2,520/2,520	11.67	22.80	89.80
`170105`	all eight	2,520/2,520	10.50	20.15	94.88

4.4 Object Models, Retargeting and Physics Replay

The original object-quality gate accepted 44 of the 73 prepared episodes. Independent scratch rebuilds with two-view track recovery raised acceptance to 70 of 73, recovering 26 previously rejected episodes without changing canonical scene files. Three remained rejected. On these 70 accepted models, the median manipulated-object surfel extent along its three object axes was $46.2 \times 47.7 \times 49.7$ mm; 35 of 70 had at least one axis above 50 mm, although the nominal cube edge is 40 mm. A partially visible surfel shell is not a measured solid-cube bounding box, but the discrepancy warns against treating the model as exact grasp or collision geometry. Cube yaw is also symmetry-ambiguous without a marked face.

Forty canonical `plan.h5` files exist: 38 older-mode and two later v3 plans. Their estimated-target position RMSE has median 6.2 mm, p90 7.7 mm and maximum 122.0 mm. The 122.0 mm plan is an explicit quality failure, not evidence of usable IK. The 26 newly recovered object models were retargeted in a separate scratch branch: 22 passed its geometric plan gate, two failed retargeting and two failed plan quality. Canonical and scratch counts have different modes and gates and are not added into one yield rate.

MuJoCo replayed all 24 produced scratch plans with free cube bodies and a 50 Hz held-command controller. Under the provisional contact model, threefold-slowed commands and a simulator-only 0.10 rad grasp-pressure command, 18 of 24 passed the released, stable-stack, grasp-integrity and contact-physics checks. Among the 22 geometrically accepted plans, 17 also passed replay; one geometrically rejected plan happened to stack in this simulator but remains rejected for large target error. Robot contact with the red support occurred in all 24 replays and green-support contact in two. Thus "physics-successful" means stable stacking under these *specific* assumptions, not collision-free, physically safe or real-robot-valid motion. Static old-UR10 wrist forward kinematics agreed between Pinocchio and MuJoCo to 4.9×10^{-16} m on 15 saved poses, and the gripper pad-gap models agreed within 2.6×10^{-9} m on 15 commands. These are checks between software representations of the same robot, not hardware accuracy.

TABLE III
Object, Plan and Simulation Outcomes (Distinct Artifact Branches)

Gate or inventory	Count
Prepared episodes / recorded episodes	73 / 78
Original object models passing quality / prepared	44 / 73
Accepted with independent track-recovery scratch rebuilds	70 / 73
Canonical geometric plan files (one 122 mm RMSE outlier)	40
Recovered scratch plans passing geometric gate / attempted	22 / 26
Produced scratch plans passing MuJoCo stack check / replayed	18 / 24
Passing both geometric gate and MuJoCo stack check	17 / 22
Robot–red / robot–green support contact among scratch replays	24 / 24; 2 / 24

4.5 Local Layout Transfer and Learning

One successful recovered human-motion plan, `2026-10-06_11-01-57`, seeded local object-relative variations. Red and green cube XY positions were shifted independently; the blue cube began on red. Full-path IK and the same MuJoCo replay gate accepted 46 of 48 sampled layouts at ± 20 mm per axis, and 27 of 48 at ± 50 mm. These are *candidate generation yields* in two different random sweeps, not source-diverse human demonstrations or uniform reachability guarantees. Every accepted sweep still touched the red support. The 73 successful teacher plans from these sweeps were split by layout (51 train, 8 validation, 14 held-out test), but all derive from the *same* human episode and simulator.

A first, plain 53-dimensional state behavioural-cloning policy was trained on 17 separate physics-successful source replays, with an episode-level 12/2/3 train/validation/test split. It achieved *zero* released stacks in 17 closed-loop evaluations and did not lift the blue cube. Its low offline action-prediction error did not translate into task success. The later layout-conditioned model is deliberately narrower: it observes normalized time and the *initial*, simulator-provided red/green XY offsets, and predicts six UR10 joint-command corrections to the single source trajectory. Gripper commands, timing, contact model and source reference are fixed. It does not observe images or react to subsequent object motion, and it is supervised imitation, not reinforcement learning.

On 14 held-out *teacher-success-selected* layouts, the learned correction passed 14/14 replays versus 4/14 for the unchanged source commands on the same scenes. With the checkpoint frozen, a second seed sampled 30 new ± 50 mm layouts *without* teacher-success filtering: the learned controller passed 30/30 versus 0/30 for unchanged commands. The direct path-IK teacher had accepted only 18 of these 30; its 12 rejections are solver outcomes, not proofs of physical unreachability. Wilson 95% intervals for the learned success fractions are [0.785, 1.000] and [0.886, 1.000], respectively. These intervals condition on the fixed policy, simulator and layout sampling scheme; they do not quantify

between-training-seed, source-motion or dynamics variation. All 14 and all 30 learned replays still contacted red with the robot.

TABLE IV
Paired MuJoCo Stack Outcomes for the Layout-Conditioned Policy

Layout split	Learned correction	Fixed source	Teacher gate
Training layouts	51 / 51	10 / 51	Pass selected
Validation layouts	8 / 8	2 / 8	Pass selected
Held-out layouts	14 / 14	4 / 14	Pass selected
Independent seed, ± 50 mm	30 / 30	0 / 30	18 / 30 accepted

Historical figure unavailable after local workspace loss. Regenerate before submission.

Fig. 4.2. Matched-layout success for the learned correction and unchanged source commands; error bars show Wilson 95% intervals. The first test is conditioned on teacher-success selection; the second seed includes all 30 sampled layouts. Both use one source motion and one learned checkpoint.

4.6 Placement Quality and Contact Limitations

A separate *in-sample* search compared four successful variants on each of 18 teacher-accepted layouts from the independent seed. Ranking combined simulated support-contact impulse, XY centre error and orientation errors while keeping released stack success as a hard gate. Selecting the lowest-penalty candidate per layout reduced mean blue–green centre error from 9.39 to 4.09 mm and cube-symmetry-aware relative orientation error from 17.56 to 14.38 degrees. All 18 selected variants still stacked, but mean red-contact impulse stayed at 1.222 N s and red contact occurred in every scene. This is optimization *on those same 18 layouts*, not a held-out performance estimate; the learned policy was not retrained on selected variants.

Historical figure unavailable after local workspace loss. Regenerate before submission.

Fig. 4.3. Blue–green XY centre error before and after per-layout candidate selection in 18 already accepted simulator scenes. Points below the dashed identity line improved; the selection is in-sample and does not remove red support contact.

4.7 Statistical Interpretation and Remaining Measurements

Episodes, layout trials, paired frames and joint slots have different units and denominators. Frame-level counts are not independent camera-calibration replicates. All policy comparisons above use

matched layouts and the same low-level controller, but there is one selected motion source, one trained checkpoint and one provisional physics model. No data-free trained RL policy, visual policy, source-disjoint test, real UR10 evaluation or independently measured hand/object ground truth is available. Consequently this chapter establishes a constrained simulation proof of concept for one task and a strong failure diagnostic for support-object clearance; it does *not* establish general-purpose learning, collision-free demonstrations or Real-to-Sim-to-Real transfer.

Chapter 5

Analysis and Discussion

5.1 What the Results Establish

The 78-episode audit demonstrates an operational camera-to-plan route with visible attrition. Five recordings fail the articulated hand gate. Of 73 prepared episodes, the original object gate accepts 44 and an independent track-recovery scratch procedure accepts 70. The 22/26 geometric retarget result applies only to the previously rejected and then recovered subset; the 40 canonical plan files belong to a different branch. Adding these counts into one apparently homogeneous end-to-end percentage would misrepresent their selection and different planner modes.

Dynamic execution is no longer merely prospective. Seventeen of 22 geometrically accepted recovered plans produce released stable stacks in the tested MuJoCo model. Yet the robot hits the red support in every one of the 24 replayed scratch scenes. Consequently the simulation result is evidence for a task predicate under provisional contacts, not for a safe collision-free trajectory oracle. The 122 mm canonical target-RMSE outlier also shows why solver convergence alone is not a plan-quality gate.

The failed plain BC experiment matters: 17 physics-successful source rollouts and low offline action loss did not yield a single closed-loop stack. The later 14/14 and 30/30 outcomes are for a *different*, phase- and initial-layout-conditioned policy that retains a successful source command trajectory. It cannot be described as the same BC method with a larger dataset. The one source plan itself stacked in 22/22 near-fixed layouts, exposing how little variation those original scenes provided. The learned correction has a measurable advantage over unchanged commands on shifted layouts, not over a newly designed geometric controller or a separately trained no-ViKi model.

5.2 Error Sources and Reference Frames

The stage metrics use different references and cannot be added as an absolute geometric error budget. The separate HO-Cap pilot obtained 10.50–11.67 mm hand-joint MPJPE on two imported

sequences, but its camera pair was selected on those same frames and its semi-automatic MANO labels do not measure local Kinect calibration, object orientation or robot execution:

1. *Camera geometry.* The large ChArUco board restored visually useful cloud overlap, but neither a green automatic validation report nor cloud nearest-neighbour agreement proves absolute calibration accuracy. Board scale, coverage, depth calibration and near-field finger returns remain potential biases.

2. *Temporal pairing.* The 155 μs median Kinect device-timestamp gap is small for this cohort, but exposure timing, RGB–depth offsets and rolling object motion may still cause moving-hand disagreement. Host-arrival drift is a separate metric.

3. *Perception.* Solved-joint and supported-frame fractions measure coverage, not position accuracy. Object identity and track recovery decide which frames enter the downstream sample. The $46.2 \times 47.7 \times 49.7$ mm median surfel extents for a nominal 40 mm cube are a concrete model-geometry warning, though surfel bounds are not true solid dimensions.

4. *Retargeting.* Position RMSE compares a robot plan to estimated targets generated within the same pipeline. It does not validate the human trajectory or the true cube pose. The red support is not enforced as an obstacle in the current IK.

5. *Physical replay.* Pad compliance, friction, cube mass, command slowdown and the simulator-only grasp-pressure bias affect pickup and release. The exceptionally small wrist-FK and gripper-gap discrepancies only verify agreement between software models.

Errors need not be independent. Correlated hand and cube offsets can partly cancel in an object-relative grasp, while an oversized model can bias both grasp side and obstacle clearance. An external calibration target at table and carry heights, and a marked asymmetric object for orientation, would provide independent checks that the current data cannot supply.

5.3 Attribution and Statistical Validity

The 14-layout held-out comparison is paired, but the layouts were sampled only from teacher-successful cases. The 30-layout independent-seed test instead includes all samples in its specified ± 50 mm range, including 12 that the direct path-IK generator rejected. This is a useful robustness check on the *frozen* learned checkpoint, not evidence that the rejected layouts are physically unreachable. Wilson intervals describe uncertainty from finite layout trials under the chosen sampling distribution; they do not include training-seed, source-demonstration or physics-model variation.

Every one of the 73 teacher layouts shares the same human motion and simulator. Layout-disjoint splitting prevents literal leakage of a target layout into training but cannot test source-motion generalization. The time-conditioned model knows the task phase and source reference. It reads simulator-truth cube positions once at the start, so it does not solve visual recognition or closed-loop recovery

after disturbances. A fixed template is a meaningful no-learning comparator, but it is not a trained no-demonstration baseline. No causal claim about the benefit of ViKi relative to RL from scratch is supported.

The 18-layout placement-quality search is explicitly in-sample: variants are selected and assessed on the same scenes. The mean centre error improvement is real within those saved replays, but it is not a generalization estimate. Its unchanged 1.222 N s mean red-contact impulse and universal red contact show that better placement alone does not fix pickup clearance. The success predicate must be presented with these diagnostics, not as a collision-free success rate.

5.4 Scope and Implications

For the narrow cube task, passive RGB-D capture can lead to simulated stable-stack examples and a learned local correction that outperforms unchanged source commands on new placements. This is the completed proof-of-concept claim. It neither establishes a generally useful trajectory oracle nor a policy deployable on a UR10. Simulation contacts and executed states have not been validated against hardware; no Real-to-Sim-to-Real transfer is claimed.

The most valuable additional recordings would be *measurement-only* controls rather than more repetitions to train the current model: held-out large-board poses across the task height, static known geometry at both camera ranges, a marked cube or asymmetric object for orientation, and disjoint human trials to expose source dependence. These controls could quantify errors without motion capture, but they cannot turn self-consistency into millimetre ground truth by assertion. Pouring is outside the study scope.

Chapter 6

Conclusion

6.1 Summary

ViKi converts passive two-view RGB-D recordings of a human cube transfer into separate hand, object and proposed UR10 trajectory artifacts. A geometric plan is not itself a robot demonstration; dynamic replay with free objects supplies the stricter simulation test. This thesis reports the measured chain on 78 recordings and a local layout-conditioned learning experiment rather than promising a later unmeasured result.

6.2 Answers to the Research Questions

RQ1: pipeline yield. Seventy-three of 78 recordings pass the articulated hand gate; 44 of those pass the original object-quality gate and 70 pass after independent scratch track recovery. Of 26 newly recovered scenes, 22 pass geometric retargeting. Seventeen of those 22 also produce released stable stacks in the provisional MuJoCo replay. The branch-specific denominators and the red-support contact in all 24 replayed scratch plans preclude one unqualified success percentage. The external HO-Cap pilot measured 10.50–11.67 mm smoothed hand-joint MPJPE against semi-automatic annotations on two short sequences using either the selected pair or all eight views; it does not ground-truth the local cube recordings or their object and robot trajectories. Its source episodes and the local cohort were lost after the pilot and must be reacquired for a reproducible final evaluation.

RQ2: local demonstration generation. One recovered human motion generated 46 successful simulator variants in a 48-layout ± 20 mm sweep and 27 in an independent 48-layout ± 50 mm sweep. All derive from that one motion and touch the red support. This is local, physics-filtered candidate generation, not arbitrary scene or robot-configuration transfer.

RQ3: learned correction. A phase- and initial-layout-conditioned model trained on 73 generated variants outperformed unchanged source commands on the same 14 held-out teacher-selected layouts (14/14 versus 4/14) and 30 separately sampled layouts (30/30 versus 0/30). The earlier plain state

BC failed in all 17 closed-loop trials. The successful model retains the source trajectory and observes privileged initial cube positions; it is not RL or a vision-based closed-loop policy.

6.3 Claim Boundary and Next Validation

The evidence supports a camera-to-simulation proof of concept for one human source trajectory, local cube-layout changes, one old-UR10/Robotiq model and one provisional contact configuration. It does not demonstrate collision-free grasping, independent 3-D orientation accuracy, source-disjoint generalization, visual-policy performance, benefit over a separately trained no-data learner or real-robot transfer. Those claims require additional tests rather than a larger number of synthetic variants of the same source motion.

A small measurement-only set with known geometry, marked orientation and independently held-out human trials is the practical next step for the thesis evidence. Support-object clearance and contact-model sensitivity need correction before treating a simulated success as a robot-ready reference. A physical UR10 evaluation would additionally require camera-to-robot registration and hardware safety validation; it is not part of the present result.

6.4 Concluding Remarks

The result is useful precisely because the stages are not conflated: recording completeness, reconstruction coverage, model acceptance, geometric IK, dynamic stack success and learning on new layouts are reported separately. The local learning result is positive; the red-support collisions and failed plain BC are equally important boundaries on what ViKi has actually established.

References

Bibliography

[1] A. Mandlekar, S. Nasiriany, B. Wen, I. Akinola, Y. Narang, L. Fan, Y. Zhu, and D. Fox, "MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations," in *Proc. 7th Conf. on Robot Learning*, PMLR 229, pp. 1820–1864, 2023. [Online]. Available: `https://proceedings.mlr.press/v229/mandlekar23a.html`

[2] Y. Qin, W. Yang, B. Huang, K. Van Wyk, H. Su, X. Wang, Y.-W. Chao, and D. Fox, "AnyTeleop: A general vision-based dexterous robot arm-hand teleoperation system," in *Proc. Robotics: Science and Systems (RSS)*, 2023. [Online]. Available: `https://arxiv.org/abs/2307.04577`

[3] X. Cheng, J. Li, S. Yang, G. Yang, and X. Wang, "Open-TeleVision: Teleoperation with immersive active visual feedback," in *Proc. Conf. on Robot Learning (CoRL)*, 2024. [Online]. Available: `https://arxiv.org/abs/2407.01512`

[4] R. Ding, Y. Qin, J. Zhu, C. Jia, S. Yang, R. Yang, X. Qi, and X. Wang, "Bunny-VisionPro: Real-time bimanual dexterous teleoperation for imitation learning," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2025. [Online]. Available: `https://arxiv.org/abs/2407.03162`

[5] S. Yang, M. Liu, Y. Qin, R. Ding, J. Li, X. Cheng, R. Yang, S. Yi, and X. Wang, "ACE: A cross-platform visual-exoskeletons system for low-cost dexterous teleoperation," in *Proc. Conf. on Robot Learning (CoRL)*, 2024. [Online]. Available: `https://arxiv.org/abs/2408.11805`

[6] **[Authors]**, "DexFlow: Dexterous retargeting with C^2 continuity via Hessian regularisation," *arXiv:2505.01083*, 2025. **[Verify author list and venue.]**

[7] **[Authors]**, "**[Title]**," 2025. **[Robust Huber-loss retargeting formulation cited in Section 2.3; confirm identifier before submission.]**

[8] **[C.]** Xin, **[Y.]** Yu, **[X.]** Jiang, **[Y.]** Zhang, and **[H.]** Li, "Analyzing key objectives in human-to-robot retargeting for dexterous manipulation," *arXiv:2506.09384*, 2025.

[9] S. Caron *et al.*, "Pink: Python inverse kinematics based on Pinocchio," software library. [Online]. Available: `https://github.com/stephane-caron/pink`

[10] Y. Zhu, **[et al.]**, "Vision-based manipulation from single human video with open-world object graphs," *arXiv:2405.20321*, 2024; *Autonomous Robots*, 2026. **[Confirm final volume and pages.]**

[11] J. Li, Y. Zhu, Y. Xie, Z. Jiang, M. Seo, G. Pavlakos, and Y. Zhu, "OKAMI: Teaching humanoid robots manipulation skills through single video imitation," in *Proc. Conf. on Robot Learning (CoRL)*, *PMLR*, vol. 270, pp. 299–317, 2024.

[12] Y. Qin, Y.-H. Wu, S. Liu, H. Jiang, R. Yang, Y. Fu, and X. Wang, "DexMV: Imitation learning for dexterous manipulation from human videos," in *Proc. European Conf. on Computer Vision (ECCV)*, 2022.

[13] **[Authors]**, "SPOT: SE(3) pose trajectory diffusion for object-centric manipulation," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2025. [Online]. Available: `https://arxiv.org/abs/2411.00965`

[14] **[Authors]**, "NovaFlow: Zero-shot manipulation via actionable flow from generated videos," *arXiv:2510.08568*, 2025.

[15] A. Chen, A. Lessing, Y. Liu, and C. Finn, "Curating demonstrations using online experience," *arXiv:2503.03707*, 2025.

[16] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *arXiv:2502.19645*, 2025.

[17] S. Kareer, D. Patel, R. Punamiya, P. Mathur, S. Cheng, C. Wang, J. Hoffman, and D. Xu, "EgoMimic: Scaling imitation learning via egocentric video," in *Proc. Conf. on Robot Learning (CoRL)*, 2024. [Online]. Available: `https://arxiv.org/abs/2410.24221`

[18] T. Tao, M. K. Srirama, J. J. Liu, K. Shaw, and D. Pathak, "DexWild: Dexterous human interactions for in-the-wild robot policies," in *Proc. Robotics: Science and Systems (RSS)*, 2025. [Online]. Available: `https://arxiv.org/abs/2505.07813`

[19] **[Authors]**, "Tool-as-Interface: Learning robot policies from observing human tool use," *arXiv:2504.04612*, 2025.

[20] **[Authors]**, "Hand-centric human-to-robot trajectory transfer from video demonstrations via open-world contact localization," *arXiv:2606.10743*, 2026. **[Preprint; peer-review status unconfirmed.]**

[21] **[Authors]**, "Video-to-robot imitation learning for bimanual dexterous manipulation from human video," in *Proc. ICEILM*, 2025, doi:10.1145/3788731.3788736. **[Verify author list from the published PDF.]**

[22] **[Authors]**, "RoboPaint: From human demonstration to any robot and any view," *arXiv:2602.05325*, 2026. **[Preprint; status unconfirmed.]**

[23] **[Authors]**, "Vision-based hand shadowing," *IEEE Access*, 2026; *arXiv:2603.11383*. **[Preprint; status unconfirmed.]**

[24] **[Authors]**, "DexCanvas: Bridging human demonstrations and robot learning for dexterous manipulation," *arXiv:2510.15786*, 2025.

[25] L. Romeo, R. Marani, A. G. Perri, and T. D'Orazio, "Microsoft Azure Kinect calibration for three-dimensional dense point clouds and reliable skeletons," *Sensors*, vol. 22, no. 13, art. 4986, 2022.

[26] W. Darwish, Q. Bolsée, and A. Munteanu, "Robust calibration of a multi-view Azure Kinect scanner based on spatial consistency," in *Proc. IEEE Int. Conf. on 3D Immersion (IC3D)*, 2020.

[27] J. Kim *et al.*, "Robust extrinsic calibration of multiple RGB-D cameras with body tracking and feature matching," *Sensors*, vol. 21, no. 3, art. 1013, 2021.

[28] S. Bu and S. Lee, "Easy to calibrate: Marker-less calibration of multiview Azure Kinect," *Computer Modeling in Engineering & Sciences*, vol. 136, no. 3, pp. 3083–3096, 2023.

[29] M. Tölgyessy, M. Dekan, Ľ. Chovanec, and P. Hubinský, "Evaluation of the Azure Kinect and its comparison to Kinect v1 and Kinect v2," *Sensors*, vol. 21, no. 2, art. 413, 2021.

[30] G. Kurillo, E. Hemingway, M.-L. Cheng, and L. Cheng, "Evaluating the accuracy of the Azure Kinect and Kinect v2," *Sensors*, vol. 22, no. 7, art. 2469, 2022.

[31] J. Wang, Q. Zhang, Y.-W. Chao, B. Wen, X. Guo, and Y. Xiang, HO-Cap: A Capture System and Dataset for 3D Reconstruction and Pose Tracking of Hand-Object Interaction, in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2025. [Online]. Available: `https://irvlutd.github.io/HOCap/`

[32] Microsoft, "Azure Kinect DK hardware specification" and "Synchronize multiple Azure Kinect DK devices," Microsoft Learn documentation. **[Add access date.]**

[33] Intel Corporation, "Intel RealSense D400 series product family datasheet" and multi-camera configuration guidance. **[Add revision and access date.]**

[34] RidgeRun, "Multi-camera interface configurations and synchronization," technical note. **[Non-peer-reviewed vendor documentation; retain only for engineering practice, not for scientific claims.]**

[35] I. Ali *et al.*, "Accuracy evaluation of hand-eye calibration techniques for vision-guided robots," *PLOS ONE*, 2022. **[Add volume and article number.]**

[36] "Image-based geometric fusion of multiple depth images using ray casting," U.S. Patent 10 818 071, 2020.

[37] J. Zeng, Y. Tong, Y. Huang, Q. Yan, W. Sun, J. Chen, and Y. Wang, "Deep surface normal estimation with hierarchical RGB-D fusion," in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2019. [Online]. Available: `https://arxiv.org/abs/1904.03405`

[38] Y. Wang *et al.*, "HRBF-Fusion: Accurate 3D reconstruction from RGB-D data using on-the-fly implicits," *ACM Transactions on Graphics*. **[Add volume, issue and year.]**

Note: entries marked in this colour require completion or verification before submission. Preprint entries dated 2026 could not be confirmed as peer-reviewed at the time of writing and are cited as preprints.

Appendix A

Notation

TABLE I
Symbols Used Throughout the Thesis

Symbol	Meaning
W	workspace frame, defined by the ChArUco anchor
C_k	frame of camera k
R	manipulator base frame
E	manipulator end-effector frame
H	human wrist frame
O	manipulated object frame
$\boldsymbol{T}_B^A$	rigid transform from frame B to frame A, element of SE(3)
$\boldsymbol{q}_t \in \mathbb{R}^n$	manipulator configuration at timestep t
$f(\boldsymbol{q}_t)$	forward kinematics of the end-effector
$\boldsymbol{e}_t$	task-space residual, (3.9)
ω_t	per-frame perception confidence weight, (3.14)
$q_{i,t}$	geometric quality of a triangulated landmark, (3.4)
$w_{i,t}^k$	per-view weight, single-view fallback, (3.5)
$v_{i,t}^k$	detector visibility score
$c_{i,t}^k$	sensor depth confidence
ρ_δ	Huber loss with transition parameter δ
$\lambda_v, \lambda_a, \lambda_r$	velocity, acceleration and posture weights
g_t	binary gripper state, (3.6)
τ	host-side monotonic timestamp, (3.2)
T	number of timesteps in an episode

Appendix B

Exported Dataset Schema

Each episode is stored as an HDF5 group with the fields listed in Table I. The layout is compatible with LeRobot conventions; fields beyond the LeRobot minimum are retained so that curation can be repeated downstream without re-recording.

TABLE I
Per-Episode Dataset Fields

Field	Shape	Description
`observation.images.*`	$T \times H \times W \times 3$	per-camera colour streams
`observation.state`	$T \times n$	recorded joint positions from replay
`action`	$T \times (n{+}1)$	commanded joint positions and gripper
`wrist_pose_world`	$T \times 4 \times 4$	$\boldsymbol{T}^{W}_{H,t}$
`wrist_pose_object`	$T \times 4 \times 4$	$\boldsymbol{T}^{O}_{H,t}$, (3.7)
`object_pose_world`	$T \times 4 \times 4$	$\boldsymbol{T}^{W}_{O,t}$
`gripper_state`	T	binary, (3.6)
`confidence`	T	ω_t, (3.14)
`replay_residual`	T	controller tracking residual during replay
`timestamp_host`	T	host monotonic timestamp, (3.2)

Appendix C

Solver Configuration

TABLE I
Weights and Solver Parameters Used in the Reported Experiments

Parameter	Value	Selection method
λ_v (velocity)	**X.XX**	**[grid search / held-out set]**
λ_a (acceleration)	**X.XX**	**[…]**
λ_r (posture)	**X.XX**	**[…]**
α (confidence sharpness)	**X.XX**	**[…]**
δ (Huber transition)	**X.XX**	**[…]**
τ_g (gripper threshold, mm)	**XX**	**[…]**
LM damping	**X.XX**	**[…]**
Recording rate (Hz)	**XX**	—
Max replay re-solutions	**X**	—

[TODO: Weight selection must be described honestly: if weights were tuned on the same episodes used for evaluation, say so and report the consequence, or re-tune on a held-out split.]